

Ahmed Mohamed Zakaria

Full Stack (.NET & Angular) Developer

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SUMMARY

Full Stack (.NET & Angular) Developer with professional experience building enterprise healthcare applications using ASP.NET Core and Angular. Skilled in developing scalable REST APIs, optimizing SQL performance with Dapper and EF Core, and building maintainable frontend applications with Angular, NgRx, and Nx Monorepo.

EDUCATION

Bachelor's Degree in Computer Science

Sep 2021 – June 2025

MTI University — Faculty of Computer Science & Artificial Intelligence | Grade: Very Good

EXPERIENCE

.NET & Angular Developer | Rosenfield Health

September 2025 – Present

Cairo, Egypt / Hybrid

- Architected and optimized scalable web applications for radiology and medical imaging workflows, supporting multiple healthcare user roles including Doctors, Radiologists, Patients, Clerks, Referral Partners, and Administrators.
- Engineered dual data-access architecture combining EF Core for domain logic and Dapper for high-throughput reporting queries; optimized complex SQL execution plans, significantly reducing API response latency under high patient data loads.
- Structured modular frontend applications in an Nx Monorepo using Angular Signals, NgRx, and RxJS Observables for real-time clinical dashboards and diagnostic reporting, ensuring proper memory management and subscription cleanup.

PROJECTS

Tawsela Kids – School Bus Fleet & Tracking Platform

[Live Demo ↗](#)

- Built real-time school bus fleet & student tracking platform with GPS geofencing, SignalR live broadcasts, and offline-first PWA.
- Engineered WebRTC audio & video calling with Coturn (STUN/TURN), real-time instant chat, and background WebPush alerts.
- Developed student absence management, delegated pickup authorization workflows, and role-based portals using Clean Architecture & CQRS.
- Implemented comprehensive unit testing with xUnit & Moq covering core domain business rules and CQRS command handlers.

Technologies: Angular, ASP.NET Core, SignalR, SQL Server, EF Core, WebRTC, Leaflet, PWA, WebPush, MediatR, xUnit, Clean Architecture.

Full-Stack Real-Time Auction System

[GitHub Repo ↗](#)

- Engineered high-concurrency bidding logic with SignalR WebSockets and RxJS countdown timers for real-time state synchronization.
- Integrated Stripe payment webhooks, JWT with Refresh Tokens, and 2FA (OTP) authentication.

Technologies: Angular, ASP.NET Core, SignalR, EF Core, Redis, Stripe, RxJS, Bootstrap, JWT & Refresh Tokens, 2FA (OTP), Clean Architecture.

Full-Stack E-Commerce System

[GitHub Repo ↗](#)

- Designed caching using SQL Server paired with Redis, delivering high-performance product filtering & catalog queries.
- Applied Clean Architecture with Specification Pattern, decoupling business domain logic from data infrastructure.

Technologies: Angular, ASP.NET Core, SQL Server, Redis, Entity Framework Core, Stripe, JWT, Specification Pattern, Clean Architecture.

TECHNICAL SKILLS

Backend: C#, ASP.NET Core, Dapper, Entity Framework Core, SignalR, LINQ, RESTful APIs, MVC, Swagger/OpenAPI, xUnit

Frontend: Angular, Angular Signals, Nx Monorepo, NgRx, RxJS & Observables, ngx-translate (i18n), HTML, CSS, Bootstrap, TypeScript, ngx-bootstrap, Font Awesome, Toastr, ngx-spinner

Database & Caching: PostgreSQL, SQL Server, Redis, SQL Query Optimization, Indexing & Execution Plans

Payment Processing: Stripe

Authentication: ASP.NET Core Identity, JWT, 2FA (OTP), Role-Based Access Control (RBAC)

Logging & Resilience: Serilog (Structured Logging), NLog, Background Services

Architectural Patterns: Clean Architecture, CQRS Pattern, Factory Pattern, Strategy Pattern, Repository Pattern, Unit of Work Pattern, Specification Pattern

Soft Skills: Problem Solving, Agile/Scrum, Team Collaboration, Code Review & Technical Communication

LANGUAGES

Arabic — Native

English — Intermediate